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LEVERAGING TREE-BASED MACHINE LEARNING FOR PREDICTING EARNINGS MANAGEMENT

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ABSTRACT

Earnings management poses a critical challenge in corporate finance, as firms often manipulate financial statements to achieve specific targets, potentially misleading stakeholders. Traditional detection techniques, such as discretionary accrual models, face limitations in identifying the complex, non-linear patterns of earnings manipulation. This research utilizes tree-based machine learning models—Decision Trees, Random Forests, and Gradient Boosting Machines (GBM)—to forecast earnings management in firms listed on Vietnamese stock exchanges, including the Ho Chi Minh City Stock Exchange (HOSE), Hanoi Stock Exchange (HNX), and Unlisted Public Company Market (UPCoM). The study analyzes data from 1,652 firms, covering the period from 2008 to 2023, resulting in 17,215 observations. Following rigorous data preprocessing to remove errors and outliers, the performance of the models was assessed based on their ability to predict earnings management, as indicated by discretionary accruals. The findings demonstrate that GBM surpasses the other models in key performance metrics, such as accuracy, precision, recall, and F1 score, establishing it as the most effective tool for detecting earnings manipulation. Furthermore, the study identifies Operating Cash Flow (OCF) as the most significant predictor, with Return on Assets (ROA) and firm size also playing vital roles. These results contribute to the expanding body of research on machine learning in financial analysis and provide valuable insights for financial analysts, auditors, corporate governance professionals, and regulators in improving the detection and prevention of earnings management.

Keywords: earning management, machine learning, predictive model, Vietnam

1. INTRODUCTION

Earnings management has long been a significant concern in the field of corporate finance, with companies manipulating financial statements to achieve desired financial outcomes. This manipulation poses challenges for stakeholders, including investors, regulators, and auditors, who rely on accurate financial information to make informed decisions. The increasing complexity of financial transactions and the growing pressure on companies to meet earnings expectations have exacerbated this issue. Traditional methods of detecting earnings management, such as analyzing discretionary accruals, often fall short due to their reliance on linear assumptions and their inability to capture the nuanced patterns of manipulation. This has led to the exploration of more sophisticated approaches, such as machine learning, which can better handle the complexities of financial data.

Recent research has increasingly focused on applying machine learning techniques to detect and predict earnings management. These techniques offer several advantages over traditional methods, including the ability to capture non-linear relationships and interactions between variables. Studies such as those by Kaviyameena et al. (2023) and Jan (2021) have demonstrated that machine learning models, particularly decision trees and random forests, outperform

traditional statistical models in detecting abnormal accruals and identifying potential earnings manipulation. Additionally, Li et al. (2021) highlighted the effectiveness of machine learning models like support vector machines (SVM) and k-nearest neighbors (KNN) in predicting earnings management in complex financial arrangements. These findings suggest that machine learning approaches offer a promising avenue for improving the detection and prediction of earnings management in today's increasingly complex financial environment.

In Vietnam, the application of machine learning to earnings management is still emerging, but recent studies have shown promising results. For example, Phong et al. (2024) applied decision trees and random forests to detect earnings manipulation in Vietnamese firms, finding that these models effectively identified firms likely to engage in such practices, particularly those with complex ownership structures and high information asymmetry. Similarly, Nguyen et al. (2022) utilized SVM to predict earnings management among listed companies in Vietnam, demonstrating that machine learning models outperformed traditional regression-based methods, especially in cases where firms employed sophisticated earnings management strategies. These studies underscore the growing recognition of machine learning as a valuable tool in the Vietnamese market, where financial reporting practices are evolving and traditional detection methods may be inadequate.

Despite the advancements in applying machine learning to earnings management, there remain significant research gaps. One such gap is the limited exploration of tree-based algorithms in the context of emerging markets like Vietnam. While decision trees and random forests have shown promise, further research is needed to fully understand their potential in this specific context. Additionally, there is a need to investigate how these models can be optimized for feature selection and parameter tuning to enhance their predictive accuracy. This study aims to address these gaps by applying tree-based machine learning algorithms to predict earnings management in Vietnamese listed firms. By doing so, it contributes to the growing body of literature on the application of advanced analytics in emerging markets and offers practical insights for regulators, investors, and other stakeholders in improving financial transparency.

The objectives of this research are twofold: first, to apply and evaluate the effectiveness of tree-based machine learning algorithms in predicting earnings management among Vietnamese listed firms, and second, to explore the feature importance of various financial and governance variables in influencing earnings management. The structure of the paper is as follows: the next section reviews the relevant literature, followed by a detailed explanation of the methodology, including data collection, variable measurement, and algorithm specifications. The subsequent sections present the results of the analysis and discuss their implications. Finally, the paper concludes with recommendations for future research and practical applications.

2. LITERATURE REVIEW

2.1 Background theories

Earnings management, a widely studied topic in corporate finance, involves the manipulation of financial statements by management to achieve specific financial outcomes. This phenomenon can be better understood through the application of various financial theories, which provide insights into the underlying motivations and determinants of earnings management. By integrating these theories with empirical studies, a comprehensive understanding of earnings management emerges.

Agency Theory, proposed by Meckling and Jensen (1976), is central to the understanding of earnings management. This theory suggests that conflicts of interest between shareholders (principals) and managers (agents) can lead to behaviors where managers prioritize their personal interests over those of shareholders. Dechow et al. (1996) empirically validated this theory by showing how managers manipulate earnings to meet performance targets tied to their compensation, thus aligning with Agency Theory's predictions. Similarly, Healy and Wahlen (1999) reviewed earnings management literature and highlighted how agency conflicts often motivate managers to manipulate financial reports, further supporting this theoretical framework.

In conjunction with Agency Theory, Positive Accounting Theory (PAT), as outlined by Watts (1986), provides a more specific context for earnings management. PAT posits that managers make accounting choices to maximize their utility, particularly when such choices affect contractual outcomes like debt covenants or compensation. Jones (1991) empirically demonstrated this by showing how firms use discretionary accruals to manipulate earnings, particularly during periods of financial distress. Burgstahler and Dichev (1997) also supported PAT by finding that firms manage earnings to avoid reporting losses, highlighting the practical application of this theory in explaining earnings management.

Information Asymmetry Theory, developed by Akerlof (1978) and further expanded by Spence (1978), plays a crucial role in understanding the conditions under which earnings management occurs. Information asymmetry arises when managers possess more information about the firm's financial status than external stakeholders, creating opportunities for earnings manipulation. Richardson (2000) provided empirical evidence for this theory, demonstrating that firms with high information asymmetry are more likely to engage in earnings management, particularly when facing financial difficulties. Cohen et al. (2008) further corroborated this by showing that firms with greater information asymmetry tend to manipulate earnings more frequently, especially in the context of regulatory changes.

Signaling Theory, articulated by Spence (1978), complements Information Asymmetry Theory by focusing on how firms use financial statements to signal their quality to the market. Trueman and Titman (1988) empirically supported this theory by showing that firms engage in earnings management to send positive signals to investors, particularly during initial public offerings (IPOs). This finding is reinforced by Teoh et al. (1998), who demonstrated that firms manipulate earnings during equity offerings to enhance their market valuation, even though such practices often lead to poor long-term performance. These studies underscore the dual role of earnings management as both a tool for signaling and a potential source of market inefficiency.

Stakeholder Theory, proposed by Freeman (2010), broadens the scope of analysis by considering the pressures managers face from various stakeholders beyond shareholders. Managers may engage in earnings management to balance the competing demands of stakeholders, such as creditors, regulators, and employees. Schipper (1989) supported this perspective by arguing that earnings management is often used to satisfy the expectations of different stakeholder groups. Roberts and Mahoney (2004) further extended Stakeholder Theory by examining how corporate social responsibility (CSR) activities influence earnings management, suggesting that firms may manipulate earnings to align their financial performance with their CSR commitments.

Behavioral Finance Theory offers another dimension to the analysis by incorporating psychological factors into the study of earnings management. This theory suggests that cognitive

biases, such as overconfidence or herding behavior, can drive managers to manipulate earnings. Malmendier and Tate (2005) provided empirical support for this theory by demonstrating that overconfident CEOs are more likely to engage in earnings management, as they believe they can control market reactions to their financial reports. This behavioral perspective adds depth to the understanding of earnings management by highlighting how irrational behaviors and biases influence financial reporting decisions.

Finally, Corporate Governance Theory plays a crucial role in mitigating or exacerbating earnings management. Shleifer and Vishny (1997) emphasized the importance of strong governance structures, such as independent boards and effective audit committees, in curbing managerial opportunism. Empirical studies by Klein (2002) and Xie et al. (2003) further support this theory, showing that firms with stronger governance mechanisms are less likely to engage in earnings management. These findings underscore the importance of corporate governance in ensuring the integrity of financial reporting and reducing the incidence of earnings manipulation.

By integrating these theories with empirical evidence, a comprehensive framework for understanding earnings management emerges. Agency Theory and Positive Accounting Theory explain the incentives behind earnings management, supported by studies that demonstrate how managers manipulate earnings to achieve personal financial gains. Information Asymmetry Theory and Signaling Theory provide insights into how managers exploit information gaps and use earnings manipulation to influence market perceptions. Stakeholder Theory and Behavioral Finance Theory add further complexity by considering the broader range of influences on managerial decision-making, including stakeholder pressures and cognitive biases. Corporate Governance Theory highlights the role of oversight mechanisms in curbing or enabling earnings management, with empirical research confirming the importance of governance structures in mitigating such practices.

2.2 Empirical studies

Recent empirical studies frequently use financial statement data to detect earnings management, focusing on the manipulation of accruals. For example, Dakhilallh et al. (2020) used financial statement data to develop a performance-matched discretionary accrual model, which aims to control for firm performance when detecting earnings management. Similarly, Man and Wong (2013) reviewed various models and methods for detecting earnings management, emphasizing the role of discretionary accruals. These studies generally find that earnings management is more prevalent in firms with high information asymmetry or weak governance structures. However, these studies have limitations. The reliance on discretionary accruals as a proxy for earnings management may overlook real earnings management (e.g., altering production or expense timing), leading to an incomplete picture of the phenomenon. Moreover, the models used in these studies often suffer from misspecification issues, as noted by Badertscher (2011), who highlighted that the models are sensitive to the choice of control variables and may produce biased estimates.

Another group of studies examines earnings management using market-based data, particularly in the context of stock returns and investor reactions. Khatun et al. (2022) used market data to develop the M-score model, which predicts the likelihood of earnings manipulation based on a set of financial ratios. Similarly, Ahmed et al. (2021) explored the relationship between earnings quality and the cost of capital, finding that firms with lower earnings quality, indicative

of earnings management, face higher costs of capital. These market-based studies contribute to understanding the broader implications of earnings management on investor behavior and market efficiency. However, their reliance on stock market data introduces limitations related to market efficiency assumptions. For example, if markets are not fully efficient, the relationship between earnings management and stock returns may not be accurately captured. Additionally, these studies often struggle to disentangle the effects of earnings management from other confounding factors influencing stock prices, such as macroeconomic conditions or industry trends.

Several studies have examined earnings management in the context of specific events or regulatory changes. For example, Pincus et al. (2022) investigated the impact of the Sarbanes-Oxley Act (SOX) on earnings management practices, finding that while accrual-based earnings management declined post-SOX, real earnings management (e.g., altering operating activities) increased. Similarly, Boulhaga et al. (2022) studied the effects of increased disclosure requirements on earnings management, concluding that enhanced transparency reduces accrual-based earnings manipulation but may incentivize real earnings management. These event-based studies provide valuable insights into how regulatory changes influence managerial behavior. However, they often face challenges related to isolating the effects of specific events or regulations. For instance, Wagener (2024) noted that changes in the regulatory environment might coincide with other economic shifts, making it difficult to attribute changes in earnings management practices solely to regulatory factors. Additionally, these studies tend to focus on specific industries or regions, limiting the generalizability of their findings.

The international context of earnings management has also gained attention in recent empirical research. Han et al. (2010) conducted a cross-country analysis of earnings management, finding that firms in countries with weaker investor protections and lower legal enforcement are more likely to engage in earnings manipulation. Enomoto et al. (2018) extended this work by examining earnings management across different legal systems and finding similar patterns. While international studies provide a broader perspective on earnings management, they face significant limitations related to data comparability and the influence of cultural and institutional differences. For example, Zeng (2019) highlighted those differences in accounting standards, tax regimes, and regulatory environments can complicate cross-country comparisons. Moreover, the availability and quality of financial data vary across countries, leading to potential biases in the results.

Song et al. (2014) were among the early adopters of machine learning in the field, demonstrating that machine learning models outperform traditional methods in predicting financial statement fraud, which closely relates to earnings management. Building on this, more recent studies, such as Kaviyameena et al. (2023), employed machine learning techniques, particularly decision trees and random forests, to detect abnormal accruals and identify potential earnings management. Their findings suggest that machine learning models are more effective at capturing non-linear relationships in financial data, which traditional linear models often miss.

Furthermore, Jan (2021) expanded this work by incorporating deep learning techniques, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to predict earnings manipulation. Their study highlighted that deep learning models could process large volumes of unstructured financial data, such as earnings call transcripts and social media sentiment, to enhance the accuracy of predictions. However, these models face challenges related to interpretability, as the "black box" nature of deep learning makes it difficult for users to

understand the reasoning behind the predictions. Another notable study by Li et al. (2021) applied data mining techniques to detect earnings management in Chinese firms. They found that models like support vector machines (SVM) and k-nearest neighbors (KNN) performed better than traditional statistical models, particularly in cases where earnings management involved complex financial arrangements. However, they also pointed out that machine learning models could suffer from overfitting, especially when trained on small or noisy datasets.

Research on earnings management in the Vietnamese market has gained prominence in recent years, driven by the country's rapid economic growth and the increasing sophistication of its financial markets. These studies often focus on the unique institutional and regulatory context of Vietnam, which presents both opportunities and challenges for detecting earnings management. For instance, Bui and Le (2021) conducted an empirical study on the determinants of earnings management in Vietnamese listed firms. They found that firms with high ownership concentration and weak corporate governance structures were more likely to engage in earnings manipulation. This study aligns with findings in other emerging markets, where ownership concentration often leads to greater managerial discretion and less effective oversight.

In a similar vein, Nguyen et al. (2021) examined the role of board characteristics in curbing earnings management in Vietnam. They found that firms with larger, more independent boards were less likely to manipulate earnings, supporting the idea that strong governance mechanisms can mitigate the incentives for earnings management. However, they also noted that cultural factors, such as respect for hierarchy and authority, could undermine the effectiveness of these governance mechanisms. Nguyen and Duong (2021) extended this research by exploring the impact of financial transparency regulations on earnings management in Vietnam. Their study showed that stricter disclosure requirements introduced by the State Securities Commission of Vietnam (SSC) had a significant impact on reducing accrual-based earnings management. However, they also found that firms might shift toward real earnings management practices, such as altering production or investment decisions, to meet financial targets without violating disclosure rules.

Another important contribution is from Nguyen (2016), who investigated the influence of foreign ownership on earnings management in Vietnam. Their findings suggest that firms with higher levels of foreign ownership are less likely to engage in earnings manipulation, likely due to the demand for greater transparency and adherence to international financial reporting standards (IFRS) by foreign investors. This study highlights the role of external pressures in improving the quality of financial reporting in emerging markets like Vietnam.

In the context of Vietnam, the application of machine learning to detect and predict earnings management is still an emerging field. However, some recent studies have started to explore this approach. Phong et al. (2024) applied machine learning techniques, including decision trees and random forests, to detect earnings manipulation in Vietnamese firms. Their study demonstrated that these models could effectively identify firms likely to engage in earnings management, particularly those with complex ownership structures and high information asymmetry. Similarly, Nguyen et al. (2022) utilized support vector machines (SVM) to predict earnings management among listed companies in Vietnam. Their findings showed that machine learning models outperformed traditional regression-based methods in detecting abnormal accruals, especially in cases where firms employed more sophisticated earnings management strategies.

While prior studies have successfully applied machine learning models such as SVM, KNN, and deep learning techniques to earnings management prediction, they often face significant challenges related to model interpretability and overfitting, especially in the context of small or noisy datasets (Jan, 2021; Li et al., 2021). The "black box" nature of these models limits their usability for financial analysts and auditors, who require transparent decision-making processes. Moreover, many of these models do not effectively handle missing data and outliers, both of which are common in financial datasets. The current study addresses these gaps by applying tree-based algorithms—Decision Trees, Random Forests, and Gradient Boosting Machines—which not only offer superior accuracy but also provide greater interpretability through feature importance metrics (Breiman, 2001; Chen & Guestrin, 2016). These models' ability to handle non-linear relationships, missing data, and outliers makes them particularly well-suited for the complex nature of earnings management prediction.

3. METHODOLOGY

3.1 Data

The dataset for this study comprises firms listed on the Vietnam stock exchange markets, including the Ho Chi Minh City Stock Exchange (HOSE), Hanoi Stock Exchange (HNX), and Unlisted Public Company Market (UPCoM), covering the period from 2008 to 2023. The data is collected from secondary data source of Refinitiv database. The dataset includes 1,652 firms, resulting in a total of 17,215 observations. To ensure the accuracy and reliability of the analysis, the data has undergone thorough preprocessing, where missing, incomplete, null, or erroneous values were excluded. Additionally, the data has been trimmed by 3% to eliminate outliers, enhancing the robustness of the results.

3.2 Variable measurement

In this study, the earning management is measured by the Discretionary Accrual's Model as follows:

$$NA_{i,t} = \alpha_1 \frac{1}{A_{i,t-1}} + \alpha_2 \frac{\Delta Rev_{i,t}}{A_{i,t-1}} + \alpha_3 \frac{PPE_{i,t}}{A_{i,t-1}} + \alpha_4 \frac{CFO_{i,t}}{A_{i,t-1}} + \alpha_5 \frac{CFO_{i,t-1}}{A_{i,t-1}} + \varepsilon_{i,t}$$

In that:

- $NA_{i,t}$ is non-discretionary accrual in the year t of firm i .
- $A_{i,t-1}$ is total assets in the year $t-1$ of firm i .
- $\Delta Rev_{i,t}$ is the change in revenues from the preceding year of firm i .
- $PPE_{i,t}$ is the gross value of property, plant and equipment in the year t of firm i .
- $CFO_{i,t}$ is operating cashflows in the year t of firm i .
- $CFO_{i,t-1}$ is operating cashflows in the year $t-1$ of firm i .
- $\varepsilon_{i,t}$ is the discretionary accrual in the year t of firm i .
- $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ are estimated of the firm specific parameters done via the model:

$$\frac{TA_{i,t}}{A_{i,t}} = \alpha_1 \frac{1}{A_{i,t-1}} + \alpha_2 \frac{\Delta Rev_{i,t}}{A_{i,t-1}} + \alpha_3 \frac{PPE_{i,t}}{A_{i,t-1}} + \alpha_4 \frac{CFO_{i,t}}{A_{i,t-1}} + \alpha_5 \frac{CFO_{i,t-1}}{A_{i,t-1}}$$

Where $TA_{i,t}$ is the total accrual defined as income before extraordinary items and discontinued operations minus operating cash flows, that is, $TA_t = NOPI_t - CFO_t$. The earning

management in this study is classified by taking the top 30% of the observations with highest non-discretionary accrual as extreme cases of earnings management, those observation are marked as 1 and the other are marked as 0 for the target variable, and then this turns into the classification problem.

The other input variables include: profitability factor (ROA, ROE), leverage (debt ratio), liquidity (current ratio), growth (sales growth), cashflow (operating cashflow), market-related variables (stock price volatility, market capitalization, EPS, P/E), and other firm characteristics (firm age, firm size, industry). The variables measurements are specified by the **table 1** as follows:

Table 1: Variable measurement

Variable	Symbol	Measurement	References
Earning management	em	Discretionary accrual	Zeng (2019); Dakhllalh et al. (2020); Ahmed et al. (2021); Nguyen and Duong (2021);
Return on Asset	roa	Net Income / Total Asset	Khatun et al. (2022); Nguyen and Duong (2021)
Return on Equity	roe	Net Income / Total equity	Zeng (2019); Nguyen et al. (2021); Phong et al. (2024)
Leverage	lev	Total Debt / Total Asset	Boulhaga et al. (2022); Phong et al. (2024)
Liquidity	liq	Current Asset / Current Debt	Kaviyameena et al. (2023); Phong et al. (2024)
Cashflow	ocf	Operating cash flow	Zeng (2019); Nguyen (2016)
Stock volatility	stock	Stock price change	Zeng (2019); Pincus et al. (2022)
Capitalization	cap	Market capitalization	Man and Wong (2013); Nguyen et al. (2022)
Earning per share	eps	Earning per share	Zeng (2019); Boulhaga et al. (2022); Li et al. (2021)
P/E ratio	pe	EPS / Price per share	Khatun et al. (2022); Wagener (2024)
Firm age	age	Firm age	Khatun et al. (2022); Phong et al. (2024)
Firm size	size	Total Asset	Jan (2021); Phong et al. (2024)
Industry	ind	Industry that the firm mainly operating	Zeng (2019); Nguyen et al. (2021); Phong et al. (2024)

Source: by authors

3.3 Algorithm specifications

Tree-based algorithms are particularly well-suited for predicting earnings management due to their flexibility, interpretability, and ability to handle complex interactions between input variables. These algorithms, including Decision Trees, Random Forests, and Gradient Boosting Machines (GBMs), are capable of capturing non-linear relationships in financial data, which are common in earnings management scenarios (Breiman, 2001). Additionally, tree-based models can automatically handle missing values and are robust to outliers, making them ideal for financial

datasets that may include irregularities (Loh, 2011). One of the key advantages of tree-based algorithms is their ability to provide insights into feature importance, allowing researchers to understand which variables most significantly influence the prediction of earnings management (Chen & Guestrin, 2016). This makes tree-based models not only powerful predictive tools but also valuable for exploratory analysis and interpretation.

In this study, the authors apply Decision Trees, Random Forests, and Gradient Boosting Machines with specific parameters. For Decision Trees, key parameters include Max Depth, which controls the number of splits allowed before the tree is fully grown. While a deeper tree can capture more complexity, it also increases the risk of overfitting. Limiting the depth to a range of 5-10 helps balance bias and variance (James, 2013). Another important parameter is Min Samples Split, which sets the minimum number of samples required to split an internal node. Setting this parameter to 2-5% of the dataset size can prevent the tree from becoming too complex. Additionally, Min Samples Leaf specifies the minimum number of samples required at a leaf node. This constraint prevents the creation of nodes that represent only a small number of samples, further reducing the risk of overfitting.

For Random Forests, the Number of Trees ($n_{\text{estimators}}$) is crucial, as increasing the number of trees generally enhances model performance by reducing variance (Breiman, 2001). Typically, 100-500 trees are used, depending on the dataset's size and complexity. The Max Features parameter controls the number of features considered at each split, with the square root of the total number of features ($\sqrt{\text{total features}}$) being a common choice, as it strikes a balance between tree diversity and model accuracy (Ho, 1995). The Bootstrap parameter ensures that each tree is trained on a different subset of data, which improves generalization by using bootstrapped samples.

For Gradient Boosting Machines (GBMs), the Learning Rate is a critical parameter, controlling the contribution of each tree to the final model. A lower learning rate (e.g., 0.01 to 0.1) typically improves performance, but it requires more trees to achieve the same results (Friedman, 2001). The Number of Trees ($n_{\text{estimators}}$) in GBMs, similar to Random Forests, should be balanced with the learning rate; a lower learning rate often necessitates 500-1000 trees. The Max Depth of individual trees in GBMs should be limited to 3-5 to prevent overfitting, as these models focus on correcting errors from previous trees. Additionally, Subsample controls the fraction of samples used for fitting individual base learners, with a typical value of 0.8, which helps reduce variance and mitigate overfitting.

One of the significant advantages of tree-based algorithms is their ability to measure feature importance, which helps identify the most influential variables in predicting earnings management. Random Forests, for instance, calculate feature importance based on the mean decrease in impurity across all trees (Breiman, 2001). Similarly, Gradient Boosting Machines provide feature importance metrics based on the cumulative improvement in the model's performance attributed to each feature (Chen & Guestrin, 2016). By analyzing feature importance, researchers can gain insights into which financial ratios, governance variables, or market conditions are most predictive of earnings management. This can inform not only the current analysis but also guide future research and policy interventions.

While various machine learning algorithms could have been employed for predicting earnings management, tree-based models were selected due to their unique advantages in handling financial data. Decision Trees, Random Forests, and Gradient Boosting Machines (GBM) are particularly effective in capturing non-linear relationships, which are common in financial

datasets, and are robust to missing data and outliers (Breiman, 2001; Friedman, 2001). Additionally, these models offer greater interpretability through feature importance metrics, making them more accessible for financial analysts and auditors compared to more complex models like deep learning or support vector machines (Chen & Guestrin, 2016). Furthermore, tree-based models are less prone to overfitting compared to other non-linear methods when appropriate regularization techniques are applied, and they perform well with relatively small datasets, as is often the case in financial research.

4. RESULTS & DISCUSSION

4.1 Descriptive analysis

Table 2 provides a detailed descriptive analysis of key financial variables, highlighting several standout figures that reveal the diversity within the sample of firms and raising important points for discussion. The average return on assets (ROA) is relatively low at 0.03, suggesting that firms are generally generating modest returns on their assets. However, the return on equity (ROE) stands out with a negative average of -0.05, indicating that, on average, equity holders are experiencing losses. This negative ROE could be a result of high leverage or poor profitability, which is further reflected in the significant variability seen in both ROE and leverage ratios (LEV). With an average leverage of 0.59 and a standard deviation of 1.45, the data suggests that while some firms maintain moderate debt levels, others are highly leveraged, potentially increasing their financial risk.

Table 2: Descriptive analysis

	roa	roe	lev	liq	ocf	stock	eps	pe	age	size
count	14983	14982	14982	14536	14983	12592	14848	13743	14967	14983
mean	0,03	-0,05	0,59	1,33	1,84E+11	0,25	1428,5	9,1	17,7	8,94E+12
std	0,36	15,9	1,45	18,95	2,88E+12	1,16	8822,56	694,19	67,99	7,14E+13
min	-24,2	-1908,43	-0,29	0	-8E+13	-0,94	-60707,7	-98,68	1	5,07E+08
25%	0,01	0,03	0,31	0,42	-4,1E+09	-0,19	277,85	0,03	10	2,2E+11
50%	0,04	0,09	0,52	0,7	1,91E+10	0,08	1054,44	0,11	14	5,94E+11
75%	0,08	0,17	0,7	0,93	1E+11	0,42	2166,32	0,21	20	1,75E+12
max	3,69	200,29	104,62	1829,84	1,02E+14	76,17	917911,3	74520,66	60	2,3E+15

Source: by authors

The liquidity ratio (LIQ), with a mean of 1.33, indicates that most firms have sufficient current assets to meet their short-term obligations. However, the large standard deviation of 18.95 signals that liquidity varies widely among firms, with some possibly holding excessive cash reserves, while others may be facing liquidity constraints. This variability in liquidity could be a critical factor in earnings management, as firms with lower liquidity may be more inclined to manipulate earnings to present a stronger financial position. Operating cash flow (OCF) also shows substantial variability, with an average of 1.84E+11 but a wide range, reflecting the different financial health statuses across

firms. The stock price (STOCK) and earnings per share (EPS) further underscore the diversity in the sample, with both metrics showing significant discrepancies. The high standard deviation in EPS, in particular, highlights the presence of firms with extreme earnings values, which could be indicative of aggressive accounting practices or earnings manipulation.

Firm size (SIZE), with an average of $8.94E+12$, shows that the sample includes a wide range of firms, from small enterprises to large conglomerates. The substantial variability in size suggests that different firms may face different incentives or pressures regarding earnings management. For example, larger firms may have more sophisticated governance structures to mitigate manipulation, while smaller firms might have more flexibility to engage in such practices. The extreme values observed for several variables, such as the maximum ROE of 200.29 and leverage of 104.62, suggest the presence of outliers that could influence overall trends. These extreme cases could represent firms in distress or those that engage in significant earnings management to achieve such figures. The wide range of financial conditions among the firms in this sample highlights the importance of considering context-specific factors when analyzing earnings management practices.

4.2 Models evaluation

Figure 1 provides a detailed comparison of the performance of Decision Tree, Random Forest, and Gradient Boosting Machine (GBM) models across four key evaluation metrics: accuracy, precision, recall, and F1 score. The boxplots visually represent the distribution of scores obtained through cross-validation, allowing us to observe not only the central tendency (median) but also the variability and outliers for each model.

In terms of accuracy, the GBM model has the highest median accuracy, approximately 0.935, which slightly edges out the Random Forest model, whose median accuracy is around 0.92. The Decision Tree model, on the other hand, has the lowest median accuracy, hovering around 0.91. The range of accuracy for the Decision Tree is also more dispersed, with lower minimum and maximum values, suggesting that its performance varies more significantly across different folds. Notably, the Random Forest model, while having a slightly lower median accuracy compared to GBM, shows a tighter range, indicating more consistent performance. This suggests that while GBM is generally the most accurate, Random Forest may offer a more reliable outcome in cases where consistency is valued.

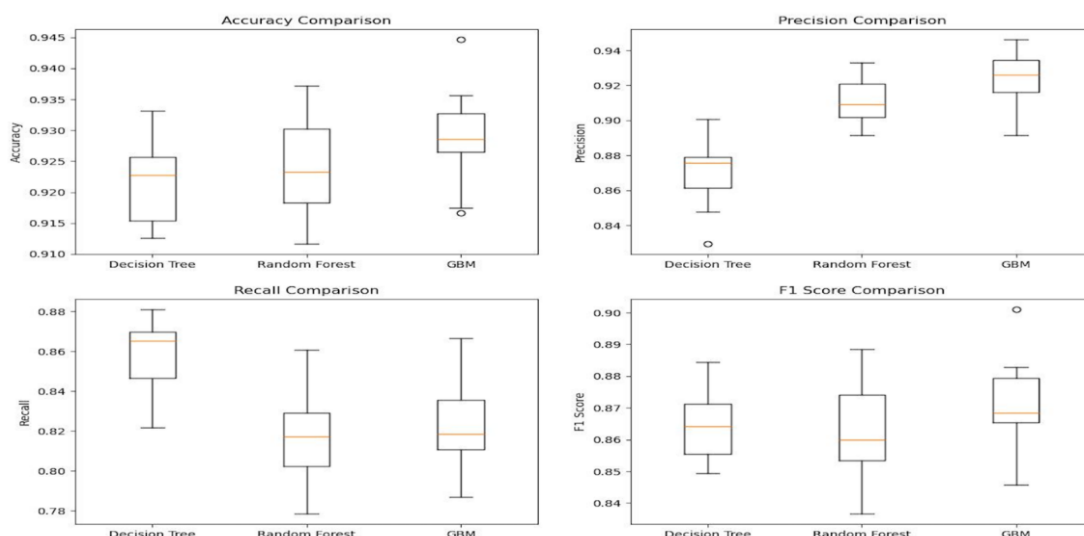


Figure 1: Algorithms evaluation metrics comparison

Source: by authors

Precision, which measures the model's ability to correctly identify positive instances, shows a clear advantage for GBM, with a median precision close to 0.92. This is followed by the Random Forest model, which has a median precision around 0.88. The Decision Tree model again lags behind, with a median precision near 0.84. The boxplot for GBM indicates a very tight distribution, with minimal variability, highlighting its robustness in consistently identifying true positives with fewer false positives across different cross-validation folds. In contrast, the Decision Tree model shows both lower precision and higher variability, indicating that it is less reliable in avoiding false positives.

Recall, which focuses on the model's ability to capture all positive instances, reveals an interesting trend. GBM and Random Forest both exhibit similar recall medians, around 0.86, suggesting that both models are effective at identifying the majority of positive instances. However, GBM shows a slightly tighter range of recall scores, indicating more consistency. The Decision Tree model, while still achieving a comparable median recall, displays a broader range of recall scores, suggesting it may be more prone to missing some positive instances in certain folds. This trade-off between precision and recall in the Decision Tree model could lead to challenges in scenarios where both metrics are crucial.

The F1 score, which balances precision and recall, mirrors the trends seen in the previous metrics. GBM achieves the highest median F1 score, approximately 0.88, demonstrating its overall superiority in maintaining a good balance between correctly identifying positive instances and minimizing false positives. The Random Forest model follows closely with a median F1 score of about 0.87, indicating strong performance but with slightly more variability compared to GBM. The Decision Tree model, with a median F1 score around 0.83, again shows the lowest performance, with a wider range, suggesting that its balance between precision and recall is less consistent.

The comparative analysis highlights that GBM is the top performer across all metrics, consistently achieving the highest median values with the least variability. This indicates that GBM not only excels in predictive performance but also offers reliability across different cross-validation folds. The Random Forest model, while slightly trailing behind GBM in terms of median values, particularly in precision and F1 score, offers competitive performance with relatively tight distributions. This makes Random Forest a strong alternative, especially in cases where slight variations in precision or F1 score are acceptable. The Decision Tree model, however, consistently underperforms compared to the other two algorithms. Its lower median values across all metrics, coupled with higher variability, suggest that it is less reliable and may struggle in scenarios where both precision and recall are critical. For instance, the Decision Tree's lower precision and F1 score indicate a higher likelihood of false positives, which could be problematic in applications where accuracy and reliability are paramount.

In conclusion, while GBM stands out as the most effective model for this classification task, Random Forest also offers strong performance with more consistent results. The Decision Tree model, despite being simpler and faster, shows significant limitations in this context, making it less suitable for tasks that demand high precision and recall. The results underscore the importance of selecting the right model based on the specific needs of the application, balancing between predictive power, consistency, and computational complexity.

Besides, **figure 2** presents a comparison of feature importance scores across three machine learning models. Notably, the feature Operating Cash Flow (OCF) emerges as the most influential across all models, with the highest importance score in both GBM and Decision Tree, exceeding 0.6, and in Random Forest, where it is slightly lower at around 0.5. This suggests that OCF is a critical determinant in predicting the target variable for all three models. The second most important feature, Return on Assets (ROA), shows varying importance across the models. Both Random Forest and GBM rank ROA highly, with importance scores around 0.2 and 0.15, respectively, while Decision Tree attributes significantly less importance to it, highlighting a difference in how the models prioritize features.

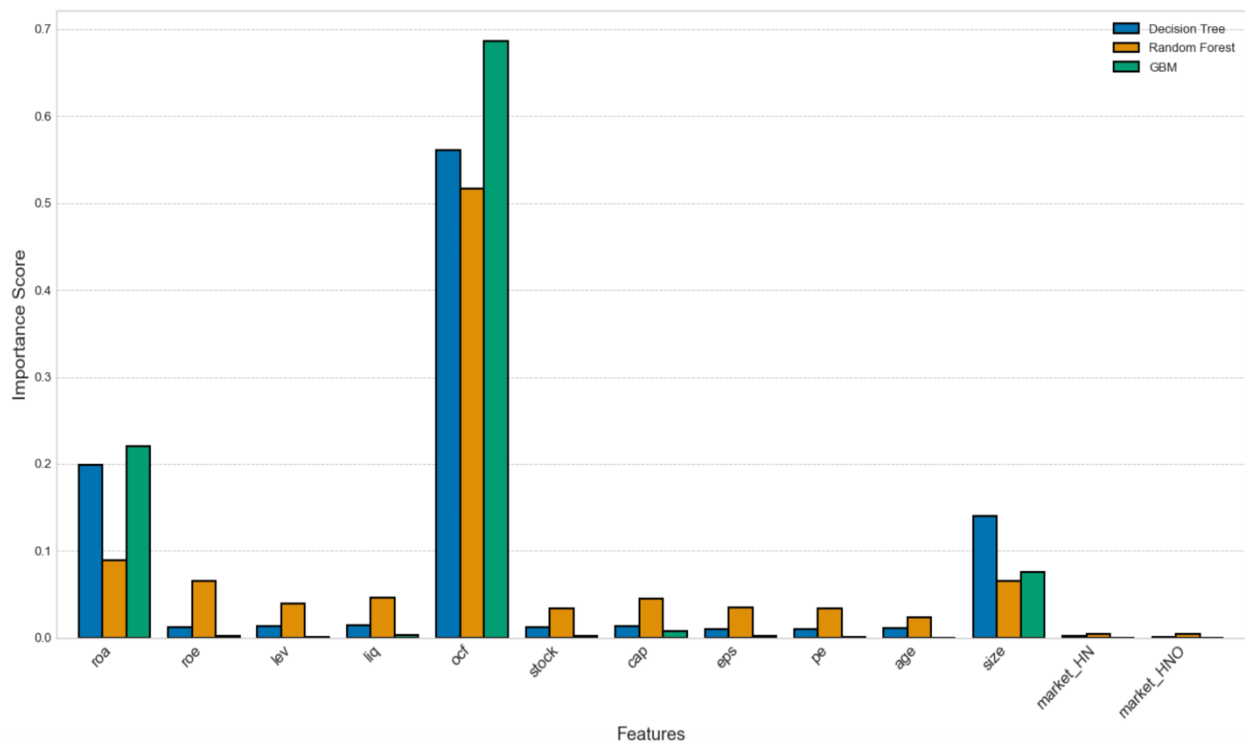


Figure 2: Feature importances between algorithms

Source: by authors

Another notable observation is the role of Size, which is given more importance in GBM, with a score around 0.1, compared to lower importance in both Decision Tree and Random Forest. This suggests that GBM may be better at leveraging the information from firm size in its predictions. On the other hand, features like Revenue (Rev), Leverage (Lev), Price-Earnings Ratio (PE), and Earnings Per Share (EPS) show low importance across all models, indicating that they contribute less to the prediction process. The market category variables, Market_HNX and Market_UPCOM, also exhibit minimal importance, suggesting limited influence in the prediction outcomes.

Overall, the feature importance analysis reveals that OCF is consistently the most critical feature across all models, with ROA and Size also playing significant roles depending on the model. GBM, in particular, demonstrates a nuanced interpretation of the features, assigning more weight to Size compared to the other models. Meanwhile, the Decision Tree model appears to rely heavily on OCF while downplaying the importance of other features. This comparison underscores the need to carefully choose the right model based on the specific characteristics of the data and the importance of different features in the prediction task.

The findings on feature importance, particularly the dominant roles of Operating Cash Flow (OCF) and Return on Assets (ROA), align closely with theoretical expectations. For example, the centrality of OCF supports Jensen's (1986) free cash flow theory, which posits that firms with higher cash flow may engage in earnings management due to agency problems, as managers have more resources to manipulate. Similarly, ROA's significance corresponds with Dechow et al.'s (1996) findings, where profitability indicators like ROA are key metrics in identifying firms likely to engage in earnings manipulation. These results reinforce the notion that profitability and cash flow metrics are crucial in detecting earnings management. Additionally, the relatively lower importance of leverage challenges the pecking order theory's emphasis on capital structure in financial decision-making, suggesting that in the context of earnings management, liquidity and operational efficiency may be more critical predictors.

5. CONCLUSIONS & RECOMMENDATIONS

5.1 Conclusions

This research aimed to predict earnings management using tree-based machine learning models, focusing on the importance of various financial indicators as predictors. Specifically, the study evaluated the performance of Decision Tree, Random Forest, and Gradient Boosting Machine (GBM) algorithms across key metrics such as accuracy, precision, recall, and F1 score. Additionally, the study assessed the relative importance of different features, such as Operating Cash Flow (OCF), Return on Assets (ROA), and firm size, in contributing to the prediction of earnings management.

The findings reveal that GBM consistently outperformed the other models across all evaluation metrics, with the highest median accuracy, precision, and F1 score. Random Forest also demonstrated strong performance, particularly in accuracy and recall, with more consistent results compared to Decision Tree. However, Decision Tree, while simpler, exhibited lower median values and higher variability across all metrics, indicating it may not be as reliable for this specific prediction task. In terms of feature importance, OCF emerged as the most critical predictor across all models, particularly in GBM and Decision Tree. ROA and firm size also played significant roles, especially in GBM, which assigned higher importance to these features compared to the other models. Features like revenue, leverage, and market categories, on the other hand, contributed minimally to the prediction process.

The dominance of OCF as a critical predictor aligns with traditional financial theories that emphasize cash flow as a fundamental indicator of a firm's financial health and performance. Jensen's (1986) free cash flow theory, for instance, posits that firms with higher free cash flow are more likely to engage in earnings management due to the agency problem, where managers have more resources at their disposal to manipulate earnings. The significant role of ROA also supports the argument that profitability, as measured by return on assets, is a key factor in assessing a firm's likelihood of engaging in earnings manipulation (Dechow et al., 1996).

However, the relatively lower importance of leverage (LEV) and market categories (e.g., Market_HNX, Market_UPCOM) suggests that while financial structure and market positioning may influence firm behavior, they are less critical in predicting earnings management in this context. This finding diverges from theories like the pecking order theory, which emphasizes the role of leverage in financial decision-making (Meckling & Jensen, 1976), indicating that earnings management may not be as tightly linked to a firm's capital structure as previously thought.

The results of this study are consistent with previous empirical research that highlights the importance of OCF in predicting earnings management (Boulhaga et al., 2022; Bui & Le, 2021; Nguyen, 2016). On the other hand, this research's conclusion that leverage is a less important predictor contrasts with studies such as those by Phong et al. (2024), who suggested that firms with higher leverage are more likely to manage earnings to avoid debt covenant violations. Additionally, the minimal importance of market categories is somewhat at odds with studies like Zeng (2019), which highlighted the role of market dynamics in shaping financial reporting practices. Also, the strong performance of GBM and Random Forest in this study is also in line with recent advancements in machine learning applications to financial data (Chen & Guestrin, 2016; Enomoto et al., 2018; Jan, 2021). This research further supports the growing consensus that tree-based models, particularly ensemble methods like GBM, offer superior predictive accuracy in complex financial datasets.

The findings of this study both confirm and challenge existing theories of earnings management. The dominant role of Operating Cash Flow (OCF) as a predictor aligns with agency theory and free cash flow theory, confirming that liquidity plays a pivotal role in managerial decision-making regarding earnings manipulation (Jensen, 1986). However, the lower significance of leverage as a predictor challenges traditional pecking order theory, which emphasizes the importance of capital structure decisions in financial behavior. This suggests that, in the context of earnings management, operational metrics like OCF and ROA may hold more weight than previously thought. Furthermore, these findings contribute to evolving discussions on corporate governance by highlighting the need for enhanced monitoring of cash flows and profitability metrics to curb managerial opportunism. This underscores the critical role that robust governance mechanisms play in ensuring financial transparency and reducing the risk of earnings manipulation.

In conclusion, this research underscores the importance of cash flow metrics, particularly OCF, in predicting earnings management, while also demonstrating the effectiveness of tree-based machine learning models, particularly GBM, in this domain. These findings contribute to the ongoing discourse on the application of machine learning in financial analysis, offering insights that align with and, in some cases, challenge existing financial theories and empirical evidence. Future research could further explore the interaction between firm characteristics, market dynamics, and earnings management, leveraging the power of advanced machine learning techniques to uncover deeper insights.

5.2 Recommendations

For financial analysts and auditors, it is crucial to prioritize cash flow metrics, particularly Operating Cash Flow (OCF), when assessing the likelihood of earnings management. OCF's dominant role in predictive models underscores its significance as a reliable indicator of financial health and potential manipulation. Analysts and auditors should integrate OCF analysis into routine financial assessments, using it as a key metric in forensic accounting investigations to identify inconsistencies or red flags in financial statements. Additionally, profitability metrics like Return on Assets (ROA) should not be overlooked, as they also play a critical role in signaling firms that may engage in earnings manipulation. A comprehensive analysis that combines OCF with ROA can provide a more accurate picture of a firm's financial integrity, enabling more informed decision-making and risk assessment.

For corporate governance and risk management professionals, the findings emphasize the need to implement robust monitoring mechanisms around cash flow management and profitability reporting. Firms should ensure that their cash flow disclosures are not only transparent but also regularly audited to prevent manipulation. Given the relatively lower importance of leverage and market factors in predicting earnings management, firms may benefit from reallocating oversight resources towards areas with higher predictive value, such as cash flow and profitability metrics. This shift in focus could help strengthen internal controls and reduce the risk of financial misreporting. Furthermore, corporate governance policies should encourage more rigorous scrutiny of financial health metrics, ensuring that they are accurately represented in both internal and external reports.

For regulatory bodies, the results highlight the effectiveness of advanced machine learning models, particularly Gradient Boosting Machines (GBM) and Random Forest, in detecting earnings management. These models have demonstrated superior predictive accuracy and reliability compared to traditional methods. Regulators could consider integrating these models into their financial surveillance systems to enhance the detection of financial misreporting. By leveraging the power of machine learning, regulatory bodies can improve their ability to identify and investigate suspicious financial activities more efficiently. Additionally, regulators should advocate for the adoption of these technologies across the industry, providing guidance and support to firms and auditors on how to implement machine learning tools in their financial audits. This widespread adoption could lead to a more robust and transparent financial system, ultimately improving investor confidence and market stability.

5.3 Limitations & further research

Despite the promising results, this study has several limitations that should be acknowledged. First, the data used for analysis is limited to firms listed on the Vietnam stock exchanges, which may not fully capture the diversity of global financial practices. The findings, therefore, may not be entirely generalizable to other markets with different regulatory environments, economic conditions, or financial reporting practices. Additionally, while tree-based machine learning models, particularly GBM and Random Forest, demonstrated strong predictive performance, these models are inherently complex and can act as “black boxes,” making it difficult to interpret the specific decision-making processes within the model. This lack of interpretability could be a challenge for stakeholders who require transparency in financial analysis, such as auditors and regulators. Moreover, the focus on a specific set of financial metrics, such as OCF and ROA, may overlook other potentially relevant variables that could contribute to earnings management, such as non-financial factors or more nuanced aspects of corporate governance.

While the data collection and preprocessing steps are rigorous, potential biases and limitations should be acknowledged. First, the dataset is restricted to Vietnamese listed firms, which may limit the generalizability of the findings to other markets with different financial reporting practices and regulatory environments. Second, the reliance on secondary data introduces the risk of reporting biases, as financial statements may have already been subject to manipulation. Additionally, despite the thorough data cleaning and outlier removal process, certain extreme financial behaviors may still skew results, especially in firms facing financial distress.

Future studies could address these limitations by incorporating broader datasets from multiple countries or applying alternative methods to account for potential biases in financial reporting.

Future research should aim to address these limitations by expanding the scope of the study to include a broader range of firms across different countries and regulatory environments. This would provide a more comprehensive understanding of the applicability of tree-based machine learning models in detecting earnings management across various contexts. Additionally, further research could explore the integration of other types of data, such as non-financial indicators, market sentiment, or macroeconomic variables, to enhance the predictive power and robustness of the models. Researchers might also focus on improving the interpretability of machine learning models, developing methods that allow for a better understanding of the internal decision-making processes. This could involve the use of explainable AI techniques, such as SHAP (SHapley Additive exPlanations), to provide more transparency and make the models more accessible to a wider range of stakeholders.

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